

3. Project Design Phase

Project Module: Visitor Management System: -

Phase Description:

This phase focuses on application design, table structure planning, workflow automation design, UI Action implementation, and email notification architecture.

This document serves as the formal milestone artifact for the ServiceNow development workflow lifecycle.

Application & Form Design: -

Custom Scoped Application:

Created a dedicated scoped application named:

Visitor Management System:

The scoped application ensures isolation from global ServiceNow components and provides a secure environment for developing custom tables, scripts, flows, and notifications.

Visitor Registration Interface:

Developed a custom Visitor form containing the following fields:

- Number (Auto Generated)
- Visitor Name
- Employee Name
- Phone Number
- Email
- Company Name
- Visit Date
- Purpose of Visit
- Status
- Checked In Time

Form Layout Configuration:

The fields were arranged logically to improve user experience and data entry efficiency.

Key design considerations:

- Mandatory visitor information displayed first.
- Employee details linked through a reference field.
- Status tracking positioned for easy monitoring.
- Check-In information placed at the end of the form.

Flow Designer Logic: -

Automated Visitor Approval Workflow:

Designed a fully automated Flow Designer process that executes whenever a visitor record is created.

Workflow Structure:

START

↓

Visitor Record Created

↓

Update Visitor Status

(Status = Approved)

↓

Send Approval Email

↓

END

Workflow Studio | Visitor Registration Flow | Flow • Visitor Management System

Visitor Registration Flow | Active

View: [Diagram] [Code] [Test] [Edit flow] [Deactivate]

TRIGGER

- Visitor Created

ACTIONS

- Update Visitor Record
- Send Email

+ Add an Action, Flow Logic, or Subflow

ERROR HANDLER ☐ If an error occurs in your flow, the actions you add here will run.

Data | Collapse All

- Flow Variables
 - Trigger - Record Created
 - Visitor Record (Record)
 - Visitor Table (Table)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
 - 1 - Update Record
 - Visitor Record (Record)
 - Visitor Table (Table)
 - Action Status (Object)
 - 2 - Send Email
 - email (Record)
 - Action Status (Object)

Read-only | Status: Published | Application: Visitor Management System | 0

Flow Components: -

Trigger:

Record Created

Table:

Visitor

Action 1:

Update Record

Automatically updates visitor status from:

Requested → Approved

Action 2:

Send Email

Sends an automated email notification to the visitor confirming registration approval.

Client Script Design: -

Phone Number Validation:

A Client Script was implemented to improve data quality.

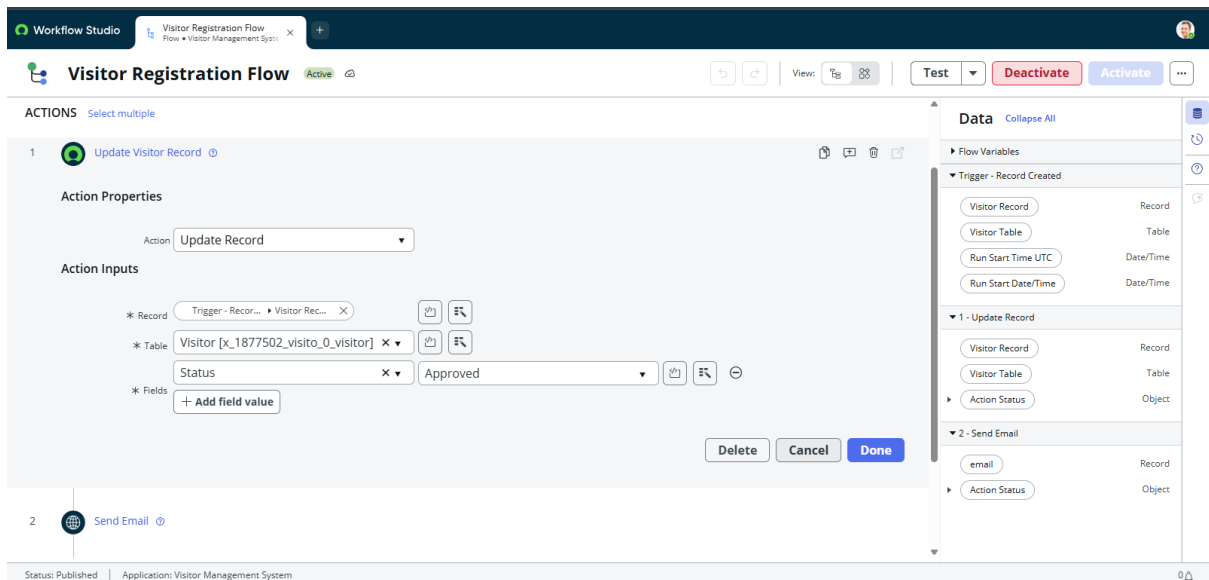
Validation Logic:

- Phone number must contain exactly 10 digits.

- Invalid entries display an error message.
- Submission is prevented until a valid number is entered.

Benefits:

- Prevents inaccurate contact information.
- Reduces manual correction effort.
- Improves visitor communication reliability.



UI Action Design: -

Check In Button:

A custom UI Action named:

Check In:

was developed for reception/security personnel.

Button Logic:

When clicked:

1. Status changes to Checked In.
2. Current date and time are automatically recorded.
3. Success message is displayed.
4. Visitor record is updated instantly.

Workflow Structure:

Approved Visitor

↓

Check In Button Clicked

↓

Status Updated to Checked In

↓

Check-In Time Recorded

↓

Success Message Displayed

↓

END

Email Notification Design: -

Visitor Approval Email:

The Flow Designer Send Email action was configured to send notifications automatically.

Subject:

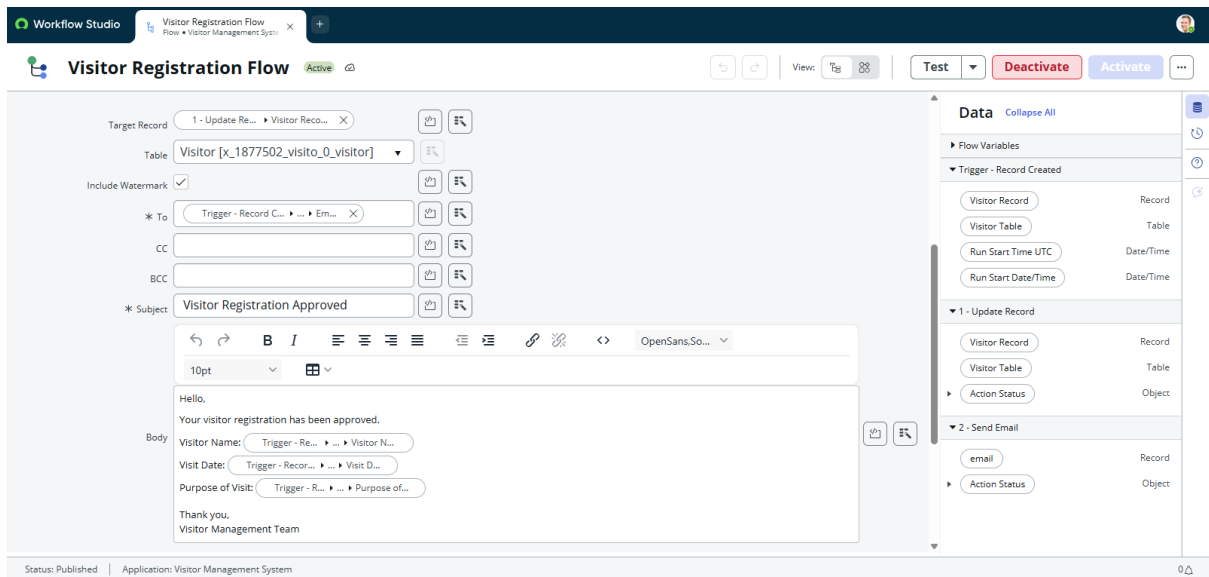
Visitor Registration Approved

Email Content:

- Visitor Name
- Visit Date
- Purpose of Visit
- Approval Confirmation

Notification Benefits:

- Immediate communication
- Reduced manual follow-up
- Improved visitor experience



Data Dictionary & Data Mapping Schema: -

Field Name	Data Type	Purpose
Number	Auto Number	Generates a unique visitor ID.
Visitor Name	String	Stores visitor identity.
Employee Name Reference	String	Links visitor to employee being visited.
Phone Number	String	Stores visitor contact number.
Email	String	Stores visitor email for notifications.
Company Name	String	Records visitor organization details.
Visit Date	Date	Stores planned visit date.
Purpose of Visit	Choice	Categorizes visit type.
Status	Choice	Tracks visitor lifecycle.
Checked In Time	Date/Time	Records actual check-in timestamp.

Design Outcome:

The Visitor Management System successfully integrates custom tables, Flow Designer automation, Client Scripts, Email Notifications, and UI Actions to provide a streamlined visitor registration and tracking solution. The design ensures data accuracy, automated processing, real-time status updates, and improved visitor management efficiency.